

Control work 9, and work on the mistakes

Linear equations and the problems behind them — Quarter III closed.

LESSON 129–130

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 9

S8 2.5 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Solve linear equations with brackets and with fractions.
- Interpret the cases of no roots and of every number being a root.
- Turn a worded problem into an equation and answer the question asked.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 358–386
Cambridge	Stage 8, pp. 31–36
Workbook	Control paper M9

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Solve $4x - 7 = 2x + 9$	3	L123–124
2	Solve $3(x + 2) = 5x - 8$	4	L123–124
3	Solve $\frac{x}{2} + \frac{x}{5} = 7$	4	L123–124
4	Solve $2(x + 4) = 2x + 8$ and say how many roots it has	4	L122–124
5	Solve $5x - 3 = 2x + 9$ naming al-jabr and al-muqabala	4	L125–126
6	Two numbers differ by 9 and add to 45: find them	6	L127–128

WHERE THE MARKS ACTUALLY GO

Q3 carries two marks for multiplying the 7 by 10 as well as the fractions; Q4 two for the words “every number is a root”; Q6 one for “let x be the smaller number”, two for the equation and one for answering with both numbers.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
sign not changed on transposing	$4x + 2x = 9 - 7$	$4x - 2x = 9 + 7$
bracket expanded partly	$3x + 2 = 5x - 8$	$3x + 6 = 5x - 8$
whole-number term not multiplied	$5x + 2x = 7$	$5x + 2x = 70$
identity read as no roots	$0 = 0$	every number is a root
moves not named	a correct but unlabelled solution	al-jabr, then al-muqabala
unknown not declared	an equation with no “let”	“let x be the smaller number”
only one number given	16	18 and 27

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Quarter III as one map

BLOCK	THE SENTENCE
parallel lines	an angle fact proves parallelism; parallelism proves an angle fact
triangles	the angles add to 180° , and an exterior angle is the two remote ones
right triangles	the acute angles add to 90° ; the leg opposite 30° is half the hypotenuse
abridged formulae	five products worth knowing by heart, in both directions
factorising	common factor, formula, grouping, splitting — in that order
algebraic fractions	cancel factors only; add over the lowest common denominator
equations	reduce to $ax = b$, then check in the original

LOOKING FORWARD

Quarter IV opens with percentages and constructions, then coordinates, functions, systems of two equations and combinatorics — and every one of those uses the equation work of this block.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $\frac{x}{2} + \frac{x}{5} = 7$.

- ① The LCD is 10.
- ② Multiply every term: $5x + 2x = 70$.
The 7 is multiplied too.
- ③ $7x = 70$
- ④ $x = 10$. Check: $5 + 2 = 7$ ✓

ANSWER

$$x = 10$$

EXAMPLE 2 Model answer, Q4: $2(x + 4) = 2x + 8$.

- ① $2x + 8 = 2x + 8$
Expand.
- ② $0 \cdot x = 0$
- ③ True for every value of x .
- ④ It is an identity: every number is a root.

ANSWER

Every number

EXAMPLE 3 Model answer, Q6: two numbers differ by 9 and add to 45.

- ① Let x be the smaller; the larger is $x + 9$.
- ② $x + (x + 9) = 45$
- ③ $2x = 36$, so $x = 18$.
- ④ The numbers are 18 and 27. Check: $18 + 27 = 45$ ✓

ANSWER

18 and 27

03 Interactive model

Give back Q6 and ask only whether the final sentence answers the question; the algebra is usually right and the answer usually incomplete.

Quarter III in twelve questions

On parallel lines, co-interior angles:

are equal

add to 90°

add to 180°

are right

Equal alternate angles prove:

parallelism

congruence

nothing

perpendicularity

The angles of a triangle add to:

90°

180°

270°

360°

An exterior angle equals:

the adjacent angle

the two remote angles

all three

90°

In a right triangle the leg opposite 30° is:

the hypotenuse

half the hypotenuse

twice it

equal to the other leg

$(a + b)^2$ equals:

$a^2 + b^2$

$a^2 + 2ab + b^2$

$a^2 - 2ab + b^2$

$2a + 2b$

$x^2 - 49$ factorises to:

$(x - 7)^2$

$(x - 7)(x + 7)$

$(x - 49)(x + 1)$

it does not

The first factorising method to try is:

a formula

the common factor

grouping

splitting

In an algebraic fraction you may cancel:

terms

factors

anything

signs

When subtracting fractions the second numerator needs:

a bracket

a denominator

nothing

a factor

$0 \cdot x = 0$ means:

no roots

one root

every number is a root

an error

Al-jabr means:

balancing

restoring

dividing

checking

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Linear equation	Chiziqli tenglama	Линейное уравнение
Root	Ildiz	Корень
To transpose	O'tkazish	Перенести
To clear fractions	Maxrajdan qutulish	Избавиться от знаменателей
Word problem	Matnli masala	Текстовая задача
Check	Tekshirish	Проверка
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 gives:

$x = 4$

$x = 8$

$x = 1$

$x = 16$

Q2 begins by:

dividing

expanding the bracket

clearing fractions

checking

Q3 is multiplied through by:

2

5

7

10

Q4 has:

no roots

one root

every number as a root

two roots

Q5 asks you to name:

the roots

al-jabr and al-muqabala

the coefficients

the check

Q6 is answered with:

one number

both numbers

the equation

x

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $4x - 7 = 2x + 9$

ANSWER $x = 8$

2. $3(x + 2) = 5x - 8$

ANSWER $x = 7$

3. $\frac{x}{2} + \frac{x}{5} = 7$

ANSWER $x = 10$

4. $2(x + 4) = 2x + 8$

ANSWER Every number

5. $5x - 3 = 2x + 9$

ANSWER $x = 4$

6. Al-jabr means

ANSWER restoring

7. Al-muqabala means

ANSWER balancing

Medium · the standard question

7 PROBLEMS

1. Two numbers differ by 9 and add to 45

ANSWER 18 and 27

2. $2(x - 3) = 2x - 5$

ANSWER No roots

3. $\frac{x}{3} - \frac{x}{4} = 1$

ANSWER $x = 12$

4. On parallel lines, co-interior $5x$ and $4x$

ANSWER 100° and 80°

5. $x^2 - 64$

ANSWER $(x - 8)(x + 8)$

6. $\frac{x^2 - 4}{x + 2}$

ANSWER $x - 2$

7. An exterior angle of 130° , one remote angle 60° : the other

ANSWER 70°

Hard · stretch and reason

7 PROBLEMS

1. $\frac{x+2}{3} - \frac{x-1}{4} = 1$

ANSWER $x = 1$

2. A rectangle 5 cm longer than wide, perimeter 42 cm

ANSWER 8 cm by 13 cm

3. Three consecutive numbers add to 81

ANSWER 26, 27, 28

4. $3(2x - 1) - 2(3x - 2) = 1$

ANSWER Every number

5. Factorise $2x^3 - 8x$

ANSWER $2x(x - 2)(x + 2)$

6. $\frac{1}{x-2} + \frac{1}{x+2}$

ANSWER $\frac{2x}{x^2 - 4}$

7. A right triangle with hypotenuse 20 and an angle of 30°

ANSWER The opposite leg is 10

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Solve $5x - 4 = 3x + 10$ and check your root.
3. Solve $\frac{x}{4} + \frac{x}{6} = 5$.

4. Two numbers differ by 11 and add to 47 . Find them.
5. Write one sentence for each of the seven blocks on the Quarter III map.